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%% Kemalettin KARA - MKT 5106 - Final Exam
clc; clear; close all;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Question 2 - Part A: System Approximation Analysis
n = [10 30];
d = conv([1 0.4 8.96], [1 1]);
Tcl = tf(n, d);

na = 30;
da = [1 0.4 8.96];
Ta = tf(na, da);

figure(1);
step(Tcl, 'b', Ta, 'r--');
legend('3rd Order System (Tcl)', '2nd Order Approximated (Ta)');
grid on;
title('Q2 Part A: Comparing Approximation');

disp('--- Part A Results ---');
stepinfo(Tcl)
stepinfo(Ta)

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Question 2 - Part B: Root Locus Controller Design (C1)
% Define the Plant (G)
num_G = 10 * [1 3];
den_G = conv([1 0.4 8.96], [1 1]);
G = tf(num_G, den_G);

% Design the Controller (C1)
K = 0.0253;
num_C = K * [1 0.4 8.96]; % Zeros canceling plant poles
den_C = [1 0];            % Integrator at s=0
C1 = tf(num_C, den_C);

% Closed Loop System (T)
L = series(C1, G);    % Open Loop
T = feedback(L, 1);   % Closed Loop

% Plot Step Response
figure(2);
step(T);
title('Q2 Part B: Closed Loop Step Response with C1');
grid on;

% Performance Characteristics
info = stepinfo(T);
fprintf('\n--- Part B Results ---\n');
fprintf('Settling Time: %.4f s (Target < 6.67 s)\n', info.SettlingTime);
fprintf('Overshoot: %.2f %% (Target < 5 %%)\n', info.Overshoot);
fprintf('Steady State Error: %.4f\n', 1 - dcgain(T));

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Question 2 - Part C: State Feedback Controller (C2)
% Define Plant Transfer Function
num = [10 30];
den = conv([1 0.4 8.96], [1 1]);
G = tf(num, den);

% Convert to State Space
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[A, B, C, D] = tf2ss(num, den);

% Check Controllability
Co = ctrb(A, B);
fprintf('\n--- Part C Results ---\n');
if rank(Co) == length(A)
    disp('System is Controllable');
else
    disp('System is NOT Controllable');
end

% Define Desired Poles
p1 = -0.6 + 0.612j;
p2 = -0.6 - 0.612j;
p3 = -10;
poles = [p1 p2 p3];

% Calculate K using Ackermann's formula
K_sf = acker(A, B, poles);
disp('State Feedback Gain K:');
disp(K_sf);

% Design Pre-scaling factor Nbar for Zero Steady State Error
DC_Gain_CL = -C * inv(A - B*K_sf) * B;
Nbar = 1 / DC_Gain_CL;
disp('Pre-scaling Factor Nbar:');
disp(Nbar);

% Create Closed Loop State Space System
A_cl = A - B*K_sf;
B_cl = B * Nbar;
C_cl = C;
D_cl = D;
sys_cl = ss(A_cl, B_cl, C_cl, D_cl);

% Plot Step Response
figure(3);
step(sys_cl);
title('Q2 Part C: Step Response of State Feedback Controller (C2)');
grid on;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Question 2 - Part D: Observer Design (C3)
% Define System
A = [0 1 0; 0 0 1; -8.96 -9.36 -1.4];
B = [0; 0; 1];
C = [1 0 0];

% Controller Gains (K) - From Part C (or analytical)
K_sf = [-1.616, 3.374, 9.8];

% Observer Design (C3) - Analytical Calculation
L = [16.9; 78.6; -50.264];

fprintf('\n--- Part D Results ---\n');
fprintf('Analytically Calculated L:\n');
disp(L);

% Verification with Ackermann
L_acker = acker(A', C', [-6 -6.1 -6.2])';
fprintf('Matlab acker() result for comparison:\n');
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disp(L_acker);

% Nbar Calculation
Nbar = 1 / (-C * inv(A - B*K_sf) * B);
fprintf('Nbar: %.4f\n', Nbar);

% Simulation (Combined System)
bigA = [ A, -B*K_sf; L*C, A - B*K_sf - L*C ];
bigB = [ B*Nbar; B*Nbar ];
bigC = [ C, zeros(1,3) ];
sys_cl_obs = ss(bigA, bigB, bigC, 0);

figure(4);
step(sys_cl_obs);
title('Q2 Part D: Step Response with Observer (C3)');
grid on;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% Question 2 - Part F: Digital Controller Verification
% Define the Plant (G)
num_G = 10 * [1 3];
den_G = conv([1 0.4 8.96], [1 1]);
G = tf(num_G, den_G);

% Define Controller C1
K_c1 = 0.0253;
num_C = K_c1 * [1 0.4 8.96];
den_C = [1 0];
C1 = tf(num_C, den_C);

% Discretize Controller (Tustin)
Ts = 0.05;
C1_d = c2d(C1, Ts, 'tustin');

% Difference Equation Extraction
[num_D, den_D] = tfdata(C1_d, 'v');

fprintf('\n--- Part F Results ---\n');
fprintf('Digital Controller Difference Equation (Ts = %.2f s)\n', Ts);
fprintf('Transfer Function C(z):\n');
C1_d

disp('Difference Equation Coefficients (Numerator):');
disp(num_D);
disp('Difference Equation Coefficients (Denominator):');
disp(den_D);
fprintf('\n');

% Verification Simulation
sys_cl_analog = feedback(C1 * G, 1);
G_plant_discrete = c2d(G, Ts, 'zoh');
sys_cl_digital = feedback(C1_d * G_plant_discrete, 1);

% Plot Comparison
figure(5);
step(sys_cl_analog, 'b', sys_cl_digital, 'r');
legend('Analog Controller', ['Digital Controller (Ts=' num2str(Ts) ')']);
title('Q2 Part F: Analog vs Digital Controller Comparison');
grid on;

% Performance Comparison
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stepinfo_analog = stepinfo(sys_cl_analog);
stepinfo_digital = stepinfo(sys_cl_digital);

disp('Performance Comparison:');
fprintf('Analog Settling Time: %.4f s\n', stepinfo_analog.SettlingTime);
fprintf('Digital Settling Time: %.4f s\n', stepinfo_digital.SettlingTime);
```